AI for Good Sandbox Hackathon – ARGOS Team – ARGOS Port Intelligence

Team leader: To be completed Team members: To be completedContact details: To be completed Github link: Repository URL to be confirmed

Introduction

ARGOS is an IoT and agent-assisted transport platform for monitoring container journeys, with an initial specialization in Central African port corridors. An enrolled ESP32 black box sends GPS and sensor telemetry; the platform validates and stores it, evaluates deterministic operational rules, and presents scoped alerts. A server-side agent registry exposes transport capabilities to an LLM assistant: telemetry monitoring, GPS and route context, risk assessment, local knowledge retrieval, regulatory scoping, and port intelligence.

This submission maps the implemented prototype and its evidence to the hackathon's requested AI-agent use case, ITU-T Y.3172 pipeline and AI Readiness Framework 2.0. The goal is to identify what ARGOS already demonstrates, what remains a prototype limitation, and how policy-grounded multi-agent scenarios could improve transport safety, cold-chain integrity and port coordination. The hackathon page emphasizes domain use cases, policy interaction, agent behavior under realistic error scenarios, knowledge-base quality, auditability and standards alignment. [1]

Description of the use case

Container operators often receive fragmented updates across vehicles, depots, roads and ports. A delayed position, route deviation, shock, unauthorized opening, fire signal, temperature excursion or low battery can remain unnoticed until cargo damage, theft or a missed handover has occurred. Port proximity is also easy to misinterpret when GPS points are treated as official terminal boundaries.

ARGOS associates a device with an authorized black box and container, checks payload, timestamp and sequence, persists telemetry and compares position with a planned corridor and configured geofences. IoT threshold rules, route monitoring and risk-window rules create box-scoped notifications. The LLM can call only registered capabilities permitted for the signed-in profile and selected boxes; it summarizes their results and provenance. Human operators remain responsible for dispatch, emergency response and regulatory decisions.

Use case requirement

Each node of the transport intelligence pipeline needs sufficient, time-synchronized and quality-labelled data; authenticated edge-to-server transfer; strict role/company/box authorization; sensor and route provenance; operationally meaningful thresholds; and a fallback when GPS, network, device or external AI services are unavailable. Cold-chain limits must be configured per cargo. Emergency or legal recommendations require a qualified human and current official sources.

Mapping with ITU-T Y.3172 Readiness 2.0 Report, and domain-specific policies

Pipeline node	ARGOS implementation and requirement	Readiness / policy evidence and gap
SRCSources	ESP32 GPS/sensors, box/container assignment, planned routes, geofences, port reference points and a versioned regulatory-source catalogue. Preserve timestamps, device sequence, quality and source.	Open and structured port/regulatory knowledge is a goal. Current port coordinates are indicative points; many catalogue entries remain “TO_VERIFY”. Obtain official datasets, licences, update owners and jurisdiction/version metadata.
CConnectivity	Authenticated IoT API, device enrollment, API validation, PostgreSQL persistence and Redis telemetry cache. Respect rate limits and tolerate missing/stale connectivity.	Y.3172-style connectivity/core functions are represented by API, cache and storage, not yet validated across multi-country networks, offline synchronization, availability or edge-cloud continuity.
PPPre-processing	Payload/time/sequence checks, normalization, box binding, quality metadata, route-corridor score, thresholds and geofence transition checks.	Controls support data integrity and purpose limitation. Document retention, lawful basis, minimization, data-subject rights, threat model and independent security review per deployment country.
MModels	Registered agents: iot_monitor, gps_route, risk_assessor, knowledge_retriever, transport_regulatory, port_intelligence. LLM function calls orchestrate available capabilities.	Risk and route scoring are deterministic rules/geometric scores, not trained ML probabilities. No benchmarked model or incident dataset is established. Keep outputs explainable; evaluate false positives, sensor bias and port/corridor coverage.
PPolicy / orchestration	Signed identity, role and selected-box checks; output provenance, trace IDs and decision-history records. Notification levels are computed and delivered in the box’s access scope.	Map operations to Cameroon/CEMAC and international sources only after verifying applicability. Human review is required for emergency escalation, cargo release, sanctions and legal advice.
DDistribution	Authenticated APIs, dashboard, GPS map, box history, notification panel and optional configured email delivery.	Provide accessible interfaces, clear uncertainty/freshness, accountable recipients and energy/availability monitoring. External LLM/provider outages must not disable sensor rules or alerts.
SINKUsers	Authorized owner, company, driver, client, port authority and superadmin views, bounded by role and box relationships.	Human-in-the-loop is operational policy, not yet a formal incident SOP with response times, acknowledgement, appeal, override and responsibility assignments.

Domain-specific references: the project catalogue includes Cameroon Law 2001/015 (transport profession; text/scope to verify), Law 2022/007 (national road heritage; applicability to verify), CEMAC road instruments (exact act to identify), SOLAS/ISPS and IMDG (scope/edition/party review), CMR and ISO/IEC 27001, 27701, 42001 and 31000 (status, licence, adoption or certification scope to verify). Catalogue presence does not establish compliance or applicability.

Error scenarios

Scenarios should release evidence progressively so agents receive incomplete and uneven information, and should record expected versus actual outputs. The current demo replays all points through the authenticated telemetry endpoint; staged release is the next competition-ready exercise capability.

Policy and human oversight

For each corridor, the operator should maintain an approved source pack: laws and regulations, transport and port mandates, privacy/cybersecurity rules, emergency responsibilities, relevant international instruments and current editions. The regulatory capability currently retrieves lexical candidates and marks applicability for verification; it is not a legal adviser. The knowledge-base owner must validate authority, date, jurisdiction, licence and update cycle before recommendations are operationalized.

Audit records should capture trace ID, actor, selected box, agent IDs/statuses, inputs' provenance, outputs, policy references, notification, human acknowledgement/override and final action. ARGOS records reviewable traces and source attribution; it deliberately does not retain hidden chain-of-thought. This is a privacy-preserving interpretation of transparent, reviewable decision-making.

Policy gaps and standards gaps derived from the mapping exercise

Conclusion and next actions

ARGOS demonstrates a transport-specific IoT workflow and a permission-bounded agent interface capable of turning telemetry into explainable operational alerts. The immediate competition work is to curate and publish an authorized knowledge base, implement progressive error-scenario playback with independent institutional roles, collect human-reviewed outcomes, and measure trace completeness, false-alert behavior, latency, availability and energy use. Human authorities retain control of safety, legal and cargo decisions.

Evidence boundary: demo figures are synthetic and validated only against the local API. They do not establish real-world predictive accuracy, regulatory compliance, official port entry, security certification or availability of the external LLM.

References

[1] ITU, “AI for Good Lab Hackathon: United Arab Emirates,” problem statement, participant requirements, evaluation criteria and submission guide (web page, accessed 29 Oct 2026): https://aiforgood.itu.int/event/ai-for-good-sandbox-hackathon-united-arab-emirates/

[2] ITU-T Recommendation Y.3172, “Architectural framework for machine learning in future networks including IMT-2020”: https://www.itu.int/rec/T-REC-Y.3172

[3] ITU, AI Readiness Framework 2.0 materials referenced by the hackathon page; exact report edition and country mapping to be confirmed before final submission.

[4] ARGOS project evidence: app/agents/registry.py, app/agents/specialists.py, app/modules/iot/, app/services/route_monitor_service.py, app/services/geofence_service.py, regulatory and port migrations, demo replay records. Local prototype inspection, 29 Oct 2026.

Draft adapted to the supplied submission template. Team identities, public repository URL and official country-policy pack remain to be completed by the team. Hackathon web page observed an “Applications closed” status while also listing a 10 November 2026 submission deadline; verify eligibility with organizers.